



SECTION 1: Amorphous & Crystalline Solids (Q1–15)

Q	Question	Answer	Explanation
1	Which of the following is an amorphous solid?	B) Glass	Glass lacks long-range order, unlike NaCl, diamond, quartz.
2	Crystalline solids have:	B) Long-range order	Atoms arranged in repeating patterns throughout the solid.
3	Which solid is isotropic in nature?	B) Amorphous solid	Properties are the same in all directions due to random atomic arrangement.
4	Which property is sharply defined in crystalline solids?	B) Melting point	Crystalline solids melt at a precise temperature.
5	Amorphous solids are also called:	A) Supercooled liquids	They behave like solids but have liquid-like disorder.
6	Which of the following solids has anisotropic properties?	B) NaCl crystal	Single crystals show direction-dependent properties.
7	Diamond is:	B) Covalent solid	Diamond has strong C–C covalent bonds in a 3D network.
8	Which solid shows plastic deformation easily?	A) Metallic crystal	Metals have dislocations allowing plastic deformation; glass, quartz, diamond are brittle.
9	Short-range order is exhibited by:	A) Amorphous solids	Only nearest-neighbor atomic arrangement is regular; no long-range order.
10	Polycrystalline solids are made up of:	B) Many small crystals	Aggregates of many small crystallites with different orientations.
11	Glass does not have:	D) Both A and C	Lacks long-range order and sharp melting point.

Q	Question	Answer	Explanation
12	Which of the following is a molecular solid?	A) Ice	Ice is held by hydrogen bonds; NaCl (ionic), diamond (covalent), Fe (metallic).
13	Quartz is:	B) Crystalline	Quartz (SiO_2) has long-range order in a lattice.
14	Ionic solids generally have:	B) High melting points	Strong electrostatic forces require high energy to break.
15	Diamond has coordination number:	A) 4	Each carbon atom is tetrahedrally bonded to 4 neighbors.



SECTION 2: Crystal Lattices & Unit Cells (Q16–40)

Q	Question	Answer	Explanation
16	Number of atoms in SC unit cell	A) 1	$8 \text{ corners} \times 1/8 = 1 \text{ atom}$
17	Number of atoms in BCC unit cell	B) 2	$8 \text{ corners} \times 1/8 + 1 \text{ center} = 2 \text{ atoms}$
18	Number of atoms in FCC unit cell	C) 4	$8 \text{ corners} \times 1/8 + 6 \text{ faces} \times 1/2 = 4 \text{ atoms}$
19	Coordination number in BCC	B) 8	Each atom touches 8 nearest neighbors.
20	Coordination number in FCC	D) 12	Each atom touches 12 nearest neighbors.
21	Coordination number in SC	B) 6	Each atom touches 6 neighbors (face-centered).
22	Edge length 'a' in SC in terms of radius r	B) $a = 2r$	Simple cubic: $\text{edge} = 2 \times \text{atomic radius}$
23	Edge length 'a' in BCC in terms of radius r	B) $a = 4r/\sqrt{3}$	Body diagonal = $4r \rightarrow a\sqrt{3} = 4r \rightarrow a = 4r/\sqrt{3}$

Q	Question	Answer	Explanation
24	Edge length 'a' in FCC in terms of radius r	A) $a = 2\sqrt{2} r$	Face diagonal = $4r \rightarrow a\sqrt{2} = 4r \rightarrow a = 2\sqrt{2} r$
25	Packing efficiency of SC	A) 52%	Low packing due to only 1 atom per unit cell.
26	Packing efficiency of BCC	B) 68%	Slightly denser than SC.
27	Packing efficiency of FCC	C) 74%	Close-packed cubic structure.
28	Examples of FCC metals	B) Cu, Al, Au	Typical close-packed metals.
29	Examples of BCC metals	C) Fe, Cr, W	BCC metals are less densely packed than FCC.
30	Simple cubic unit cell has	A) 1 atom per unit cell	$8 \text{ corners} \times 1/8 = 1 \text{ atom}$
31	Fractional contribution of corner atom	A) $1/8$	Each corner atom is shared by 8 cells.
32	Fractional contribution of face-centered atom	C) $1/2$	Each face atom is shared by 2 cells.
33	Tetrahedral void is formed by	B) 4 atoms	Four atoms form a tetrahedron around the void.
34	Octahedral void is formed by	B) 6 atoms	Six atoms form an octahedron around the void.
35	Number of tetrahedral voids per atom in FCC	B) 2	$4 \text{ atoms/unit cell} \rightarrow 8 \text{ tetrahedral voids} \rightarrow 2 \text{ per atom}$.
36	Number of octahedral voids per atom in FCC	A) 1	$4 \text{ octahedral voids per 4 atoms} \rightarrow 1 \text{ per atom}$.
37	Coordination number of tetrahedral void	B) 4	Surrounded by 4 atoms in a tetrahedron.
38	Coordination number of octahedral void	C) 6	Surrounded by 6 atoms in an octahedron.

Q	Question	Answer	Explanation
39	HCP structure has packing efficiency	C) 74%	Same as FCC (close-packed hexagonal).
40	In FCC, each atom is surrounded by	C) 12 nearest neighbours	Coordination number = 12.